Donor Account Management — Use Case Documentation

Project: Biomat USA Donor Management and Appointment Scheduling
Course: IS 6465 — Web Based Applications
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1. Overview

This document describes the four Donor Account Management use cases that make up the appointment CRUD set in the team proposal. Each description reflects the behavior of the existing application (the signup page and my account page, account.php). Authentication is treated as a precondition for the my account page access, so Log In is not described as a separate use case here.

ID	Use Case	CRUD	Summary
UC-B1	Create a donor account	Create	Donor account creation process by providing relavant information.
UC-B2	Viewing account information	Read	Donor views all of their own account information
UC-B3	Update account	Update	Donor changes their account information.
UC-B4	Delete account	Delete (hard)	Donors select ‘delete my account’ to have their information removed. This causes a cascading delete that removes account information and associated appointment deletion.

2. Terminology

These terms are used the same way in all four use cases.

Term	Meaning
Donor	A registered user signed in with a donor account that is linked to a donor profile.
System	The Biomat Donor Services web application.
‘Create account’ page	The create account page is located at register.php where a donor can successfully register their information for a Biomat donor account.
'My account’ page	The My account page opened from 'My account’ in the navigation bar. It holds the account form populated by user provided information.
Active donation center	A donation center marked active. Only active centers appear in the Donation center list.
Appointment status	Booked (scheduled), Cancelled (cancelled by the Donor), or Completed (set by personnel in the Staff workspace).
Upcoming booked appointment	An appointment that belongs to the Donor, has status Booked, and has a start date and time later than the current time. Only these can be updated or cancelled.
Date and time (UTC)	All appointment dates and times are entered and displayed in Coordinated Universal Time.

UC-B1: Create a donor account (Create)

Use Case ID	UC-B1
Use Case Name	Creating a donor account
Primary Actor	Donor
Supporting Actor	System (Biomat Donor Services web application)
CRUD Operation	Create
Description	The donor account creation process through the create account page
Preconditions	There are no preconditions for this create request. It's intended to be our sanitized entry point for new donors.
Trigger	The Donor selects 'Create Account' while they're unauthenticated; they proceed to fill out the form provided, clicking the 'create account' button located at the foot of the form.
Main Success Scenario	1. The donor selects 'Create account' on the unauthenticated page navigation bar. 2. The donor fills out the provided form with their information. This includes: a. First name b. Last name c. Date of birth (calendar selector provided) d. Social security number e. Email f. Phone number g. Street address h. City i. State or province j. Postal code k. Blood type l. Password (10 characters long enforced) 3. The donor selects the create account button at the foot of the form. 4. No errors are provided, and the account is created.

Exception Flows	1. The user is currently signed into another account. a. Creating an account button is hidden from authenticated users. b. Navigating directly to the ‘register.php’ page as an authenticated user redirects to the ‘index.php’ page. 2. The user has provided a birthday in the future or is a minor. a. The client-side form will allow the user to provide any date so long as it’s in the date format. b. When the user submits the form, the value is checked on the backend for ensuring the date is not in the future and they’re eighteen years of age. 3. The user forgets to provide a value or provides a malformed value during account creation. a. The client-side form has been marked with appropriate fields as required. This ensures that the user must provide information for account creation. Client-side validation is also enforced for value fields, but this is more for user convenience, so they do not need to submit the request to the server before receiving guidance. b. If a user submits the form by modifying the client side or sending a post request via another tool like curl, we have server-side validation that will reject the request with an associated error message and render the page again. This allows clients to see what fields they need to provide.
Postconditions	**Success** The new donor account is recorded in the SQL database with its associated account values. **Failure** No account record is recorded.
Business Rules	• Donors can create an account themselves. • Donors cannot create an account while authenticating as another user. • Donors cannot create an account without valid form values defined under ‘Main Success Scenario’. Point two subsections A-I.
Related Use Cases	The new account record can be read under UC-B2, modified under UC-B3, and deleted under UC-B4.

UC-B2: Viewing Account Information (Read)

Use Case ID	UC-B2
Use Case Name	Viewing account information
Primary Actor	Donor
Supporting Actor	System (Biomat Donor Services web application)
CRUD Operation	Read
Description	The Donor views a form of their provided account information they provided during registration and/or continued to provide post account creation.
Preconditions	The Donor is logged in with a donor account linked to a donor profile.
Trigger	The Donor selects ‘My Account’ on the navigation bar while currently authenticated.
Main Success Scenario	1. The Donor selects ‘My Account’ on the navigation bar as an authenticated user. 2. The system retrieves all account record information that belongs to the authenticated Donor. 3. The system provides the account information in editable fields related to use case UC-B3 for modifying their information, but a user doesn’t need to modify their information to view their information.
Exception Flows	1. The user is not currently authenticated but navigates to read their account information on ‘account.php’ page. a. The account page is not navigable through the GUI for an unauthenticated user to prevent accessing the page and showing unexpected values. b. The request structure for the account page in case of direct navigation needs a valid session cookie for a donor account. Without this session cookie, the system believes the user is unauthenticated and redirects to the login page to obtain a session cookie.
Postconditions	**Success** The Donor's information is provided in labeled fields for viewing. **Failure** The unauthenticated donor is redirected to the login page.
Business Rules	• Donors see only their account information. • Donors must be authenticated to see their information.
Related Use Cases	Extended by UC-B3 Update Account and UC-B4 Delete Account.

UC-B3: Update Account (Update)

Use Case ID	UC-B3
Use Case Name	Update Account
Primary Actor	Donor
Supporting Actor	System (Biomat Donor Services web application)
CRUD Operation	Update
Description	The Donor changes the information associated with their account record.
Preconditions	1. The donor is currently authenticated.
Trigger	The Donor clicks ‘My Account’ or navigates directly to ‘account.php’ while already authenticated.
Main Success Scenario	1. The user authenticates with the ‘Login’ page. 2. The user navigates to the ‘account.php’ directly or through the navigation bar. 3. The user can read their associated account information described in UC-B2. 4. The user modifies values in the editable fields. 5. The user provides values that meet the business requirements for the value structures. I.E. the user provides a nine numerical character value for their social security number.
Exception Flows	1. The user is not currently authenticated but navigates to read their account information on ‘account.php’ page. (Described in UC-B2 identically) a. The account page is not navigable through the GUI for an unauthenticated user to prevent accessing the page and showing unexpected values. b. The request structure for the account page in case of direct navigation needs a valid session cookie for a donor account. Without this session cookie, the system believes the user is unauthenticated and redirects to the login page to obtain a session cookie. 2. The user removes one of the required values or provides a malformed value. a. Client-side validation will quickly prompt the user to adjust or provide the associated values. b. In the event client-side validation is circumvented, server-side validation will return to the same page but with a rendered error message on how to properly submit the values in question.

Postconditions	**Success** The existing account record is updated in the database, so it reflects the modified values. **Failure** The account information remains unchanged.
Business Rules	• Only an authenticated user, with a current session cookie, can update an account value. • The authenticated donor can only update their information. • The authenticated donor can only view their information.
Related Use Cases	Extends the UC-B2 View account and the same validation rules.

UC-B4: Delete Account

Use Case ID	UC-B4
Use Case Name	Delete account
Primary Actor	Donor
Supporting Actor	System (Biomat Donor Services web application)
CRUD Operation	Delete (hard delete)
Description	The Donor has requested their information to be deleted from our systems.
Preconditions	1. The user is authenticated.
Trigger	The Donor is on the ‘My Account’ page and selects ‘Delete my account’
Main Success Scenario	1. The user authenticates on the login page. 2. The user navigates to the ‘My Account’ page. 3. The user enters their password on the ‘Delete your account’ form. 4. The user proceeds to click the ‘Delete my account’ button. 5. The user account information and associated appointments are deleted from the SQL database. 6. The user is unauthenticated and navigated to the homepage.
Exception Flows	1. The user is not authenticated when reaching the ‘My Account’ page. a. The user will be redirected to the login page for authentication and receiving a valid session cookie. 2. The user fails the secondary authentication on the ‘My Account’ page for account deletion. a. If the password is blank i. Client-side validation will prompt the user to provide their credentials. ii. Server-side validation will redirect to the same page with an associated error message. b. If the password is incorrect, the user will receive an associated error message on the ‘My Account’ page.
Postconditions	**Success** The account record and associated appointment records are deleted from the SQL database. The user is unauthenticated and redirected to the homepage. **Failure** The account and appointment records remain unchanged.
Business Rules	• The user must be authenticated to get to the deletion form. • The user must reauthenticate with their credentials. • The donor can only delete their account.

Related Use Cases	Extends the UC-B3 Update user account by allowing the user to delete the record instead from the same page.

3. Cascading Deletion

Deleting an account record (UC-B4) is a hard delete. The user's deletion of the account record proceeds to have the system perform a cascading deleting other account associated records such as appointments.